Samuel Greene

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Education

University of California, Davis **Expected June 2027**

Bachelor of Science, Computer Engineering & Science | ***GPA: 3.46*** *Davis, CA*

- **Relevant Coursework:** Circuits I-II, PHY 9A-D (Classical, Electromagnetic, and Modern Nuclear/Quantum Physics), Embedded Systems, Operating Systems, Advanced Computer Architecture, Theory of Computation, Algorithms & Data Structures, Machine Programming

Research Experience

CoreView Project — University of Washington **June 2026 – Present**

Systems Engineer *Seattle, WA*

- Built a **FastAPI** and **OpenSeadragon** web workspace for browsing high-resolution TIFF and OME-TIFF microscopy scans without downloading full-resolution images.
- Implemented an on-demand **Deep Zoom** tile pipeline with **libvips**, TIFF pyramid discovery, and region-only reads to serve JPEG tiles efficiently from large scans.
- Developed an operator workflow that validates TIFF/JP2 acquisition folders, submits OME-TIFF and virtual H&E generation jobs to a private GPU service, and tracks retries and results.
- Designed a **PostgreSQL**-backed image catalog and Azure Container Apps deployment architecture using Azure Files mounts, Docker, metadata indexing, search, and role-controlled workflows.

UC Davis Medical Center — Levenson Laboratory **Sep 2024 – June 2026**

Research Programmer *Davis, CA*

- Built **FIBI Post Processing**, an in-lab desktop application that accelerated microscopy image post-processing pipelines by more than **2**×, improving researcher throughput.
- Lead developer on **Maggie**, the primary software interface to custom microscopy hardware; owned the interactive **ROI-selection** system from UI design through backend integration.
- Integrated **OpenCV** feature detection and automated image-analysis tools, including color correction, multi-image overlays, and feature detection to reduce manual review.

Projects

Pathfinder | Hiking GPS Telemetry | *Github* **C++ | ESP32 | UART/I2C | OLED | Embedded Systems**

- Built ESP32 firmware that continuously parses NMEA GPS data over UART and renders hiking navigation state to an I2C OLED without blocking GPS ingestion.
- Designed a cooperative *millis()* scheduler for 4 Hz UI refresh, GPS-health diagnostics, rotating navigation/telemetry screens, and fault-aware stale-fix handling.
- Implemented configurable waypoint distance/bearing guidance and a GPS-noise-aware track-distance estimator that rejects implausible position jumps.
- Architecting a battery-powered handheld hiking navigator with a 200×300 e-paper display, integrated GPS antenna, offline topographic maps, and local trail information.

Bicycle Speed & GPS Tracking HUD **Embedded C | SPI/I2C | Sensor Fusion**

- Designed and built a real-time embedded heads-up display streaming speed and trip statistics via GPS/IMU sensor fusion.
- Wrote bare-metal firmware for concurrent peripheral drivers and interrupt-driven data capture under strict power and memory constraints.

Programming Language Tokenizer & Lexer | *Github* **C++ | Linux Toolchain | Makefile**

- Designed a tokenizer and lexer for a custom programming language grounded in formal **Theory of Computation** principles, using Pratt parsing techniques for extensibility.

Technical Skills

Hardware & Testing: Physical oscilloscopes, Saleae logic analyzers, schematics reading, electronic circuit theory
Embedded & Systems: C/C++, Embedded C, SPI, I2C, UART, GPIO, bare-metal MCU programming, sensor fusion
Software & Scripting: Python, Linux/Unix command-line, Git, Bash, C#, Rust, JavaScript, Java
Edge & Compute: Docker, CUDA, OpenCV, PyTorch, TensorFlow, Azure, AWS (EC2, S3), RunPod